

Econ 646 Spring 2011

Homework 6

The due date for this assignment is April 12th at the beginning of the tutorial.

1. *Planner's problem* Consider the following problem

$$\max_{\{C_t(i), N_t(i)\}_{i \in [0,1], t=0,1,2,\dots}} E_0 \left[\sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right) \right]$$

subject to

$$C_t = \left[\int_0^1 C_t(i)^{1-\frac{1}{\varepsilon}} di \right]^{\frac{\varepsilon}{\varepsilon-1}}$$

$$N_t = \int_0^1 N_t(i) di$$

$$C_t(i) = A_t N_t(i)$$

Notice that the allocation in this economy must be the first best, because the agent operates all the technologies by himself and maximizes the utility.

- (a) Solve this problem, and find optimal C_t .
 - (b) Compare what you find in (a) with the natural level of output we found in the lecture. Which is larger? What causes the difference?
2. *Employment subsidy* Now consider the following decentralized economy. Households solve the same problem as in the lecture:

$$\max E_0 \left[\sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right) \right]$$

subject to

$$\int_0^1 P_t(i) C_t(i) di + Q_t B_t = B_{t-1} + W_t N_t + T_t$$

$$C_t = \left[\int_0^1 C_t(i)^{1-\frac{1}{\varepsilon}} di \right]^{\frac{\varepsilon}{\varepsilon-1}}$$

Prices are flexible. But now, firms receive employment subsidy:

$$\max_{P_t(i)} \left[P_t(i) \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t - \frac{(1-\tau) W_t}{A_t} \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t \right]$$

i.e., a fraction τ of wage is subsidized by the government. The outlays associated with this subsidy are financed by lump-sum taxes (that is, they are already included in T_t).

- (a) Find equilibrium aggregate consumption/production level C_t .
- (b) Compare the production level in (a) with the production level you found in the planner's problem. Which is larger if $\tau = 0$? Is there any subsidy level at which the production level of this economy coincides with the one in the planner's economy?
- (c) Explain intuitions behind your answers in (b).